# CUSTOMER BEHAVIOR ANALYSIS

ALFIDO TECH INTERNSHIP • TASK SUBMISSION • EXECUTIVE SUMMARY

| | | | |
|---|---|---|---|
| Transaction records | 250,000 | Unique customers | 49,661 |
| Analysis period | 01 Jan 2020 – 13 Sep 2023 | Supplied churn label | 20.0% |
| Repeat-customer share | 96.7% | Inactive >180 days | 51.1% |

## EXECUTIVE SUMMARY

This project analyzes customer transaction behavior to identify purchase patterns, build RFM-based customer segments, review repeat-purchase retention and propose engagement actions for Alfido Tech. The primary file, **ecommerce_customer_data_large.csv**, was analyzed separately from the second CSV in the supplied ZIP; the files were not merged.

## METHOD

Data cleaning and validation were followed by customer-level Recency, Frequency and Monetary (RFM) feature engineering. Three K-means clusters were built using log-transformed and standardized RFM features. Category revenue, monthly transaction volume, customer inactivity and monthly cohort retention were also examined.

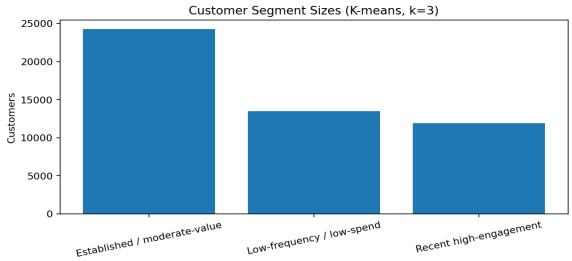
## KEY FINDINGS

The dataset includes a supplied customer-level Churn field and a substantial group of customers whose last recorded purchase was more than 180 days before the final transaction date. This inactivity group is a practical re-engagement audience, but inactivity is not treated as confirmed churn. Detailed category and segment profiles are included in the full report and notebook.

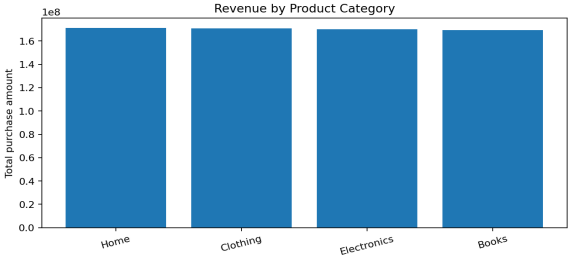
## FIVE ACTIONABLE RECOMMENDATIONS

1) Run tailored win-back journeys for customers inactive for 181+ days. 2) Encourage a second purchase with post-purchase reminders and complementary-product suggestions. 3) Reward recent high-value customers through loyalty benefits and early access. 4) Use category-specific bundles and cross-sells rather than blanket discounts. 5) Improve return-status completeness and monitor cohort retention, repeat rate and reactivation monthly.

## VISUAL SNAPSHOTS




Customer segment sizes




Revenue by product category

## DELIVERABLES & LIMITATIONS

**Notebook:** Alfido_Customer_Behavior_Analysis.ipynb — reproducible Python code and visualizations. **Full report:** Alfido_Customer_Behavior_Analysis_Report.pdf — detailed methods, segment profile, findings and recommendations. Each row is treated as a transaction because no separate transaction ID is supplied. Missing Returns values remain unknown; the churn-label definition is undocumented; clustering is descriptive, not a predictive churn model.

**Submission links:** Add the public/viewable GitHub repository URL for the notebook and README, and a viewable Google Drive URL for large assets before pasting links into the Alfido Task Submission page. External GitHub/Drive uploads have not been performed from this chat.